

0.1 calcul formule de cuadratură

10. Să se determine formula de cuadratură de grad de exactitate maxim de forma:

$$\int_{-1}^1 f(x) dx = A_1 f(-1) + A_2 f(1) + A_3 f(b) + R(f), \quad b \in [-1, 1].$$

$$R(e_i) = 0, i = \overline{0, 3}$$

unde

$$\begin{aligned} e_i &: \mathbb{R} \rightarrow \mathbb{R} \\ e_i(x) &= x^i \end{aligned}$$

adică:

$$\begin{aligned} \int_{-1}^1 dx - A_1 - A_2 - A_3 &= 0 \\ \int_{-1}^1 x dx - A_1(-1) - A_2(1) - A_3 b &= 0 \\ \int_{-1}^1 x^2 dx - A_1(-1)^2 - A_2(1)^2 - A_3 b^2 &= 0 \\ \int_{-1}^1 x^3 dx - A_1(-1)^3 - A_2(1)^3 - A_3 b^3 &= 0 \end{aligned}$$

adică:

$$\begin{cases} A_1 + A_2 + A_3 = 2 \\ A_1 - A_2 - bA_3 = 0 \\ A_1 + A_2 + A_3 b^2 = \frac{2}{3} \\ A_1 - A_2 - b^3 A_3 = 0 \end{cases}$$

scăzând ec. 2 și 4:

$$b^3 - b = 0$$

$b = 0$. Rezultă $A_1 = A_2 = \frac{1}{3}$ și $A_3 = \frac{4}{3}$. f.c. este:

$$\int_{-1}^1 f(x) dx = \frac{1}{3} (f(-1) + 4f(0) + f(1)) + R(f)$$

$$R(e_4) = \int_{-1}^1 x^4 dx - \frac{1}{3} (2) = -\frac{4}{15}.$$

$$R(f) = -\frac{4}{15} \frac{f^{(iv)}(\xi)}{4!}$$

$$\int_{-\infty}^{\infty} e^{-x^2} f(x) dx = Af(x_1) + Bf(x_2) + R(f).$$

$$R(e_i) = 0, i = \overline{0,3}$$

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-x^2} dx - A - B &= 0 \\ \int_{-\infty}^{\infty} x e^{-x^2} dx - Ax_1 - Bx_2 &= 0 \\ \int_{-\infty}^{\infty} x^2 e^{-x^2} dx - Ax_1^2 - Bx_2^2 &= 0 \\ \int_{-\infty}^{\infty} x^3 e^{-x^2} dx - Ax_1^3 - Bx_2^3 &= 0\end{aligned}$$

pot pres. că $A = B$ și $x_1 = -x_2$. Din prima: $A = B = \sqrt{\pi}/2$. Ec. a treia devine:

$$\sqrt{\pi}x_1^2 = \int_{-\infty}^{\infty} x^2 e^{-x^2} dx$$

, Solution is: $x_1 = \frac{1}{2}\sqrt{2}, x_2 = -\frac{1}{2}\sqrt{2}$.

$$R(f)=R(e_4)\frac{f^{iv}(\xi)}{4!}$$

1 Probleme cu op. de aproximare

13. Se consideră operatorul $K_n : C[0,1] \rightarrow \Pi_n$ definit prin

$$K_n(f;x) = A \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} f(x) dx.$$

- (a) Să se determine A astfel ca $K_n(e_0;x) = e_0$.
(b) Pentru A astfel determinat să se arate că

$$\lim_{n \rightarrow \infty} \|K_n f - f\|_{\infty} = 0$$

pentru orice $f \in C[0,1]$.

$$p_{n,k}(x) = \binom{n}{k} x^k (1-x)^{n-k}$$

$$\begin{aligned}
K(e_0) &= A \sum_{k=0}^n p_{n,k}(x) \int_{k/(n+1)}^{(k+1)/(n+1)} 1 dx = \\
&= A \frac{1}{n+1} \sum_{k=0}^n p_{n,k}(x) = A \frac{1}{n+1} = 1 \\
A &= n+1
\end{aligned}$$

Se fol. t. Popoviciu-Korovkin, adică e suf. să dem. că:

$$\begin{aligned}
\lim_{n \rightarrow \infty} \|K_n(e_1) - e_1\|_{\infty} &= 0 \\
\lim_{n \rightarrow \infty} \|K_n(e_2) - e_2\|_{\infty} &= 0
\end{aligned}$$

$$\begin{aligned}
K_n(e_1)(x) &= (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{k/(n+1)}^{(k+1)/(n+1)} x dx \\
&= (n+1) \sum_{k=0}^n p_{n,k}(x) \frac{1}{2} \left(\frac{(k+1)^2}{(n+1)^2} - \frac{k^2}{(n+1)^2} \right) = \\
&= \frac{1}{2} \sum_{k=0}^n p_{n,k}(x) \frac{2k}{n+1} + \frac{1}{2} \sum_{k=0}^n p_{n,k}(x) \frac{1}{n+1} = \\
&= \frac{n}{(n+1)} \sum_{k=0}^n \frac{k}{n} p_{n,k}(x) + \frac{1}{2(n+1)} 1 = \\
&= \frac{n}{n+1} B_n(e_1)(x) + \frac{1}{2(n+1)} \xrightarrow{n \rightarrow \infty} 1e_1(x) + 0
\end{aligned}$$

$$\begin{aligned}
K_n(e_2) &= (n+1) \sum_{k=0}^n p_{n,k} \int_{k/(n+1)}^{(k+1)/(n+1)} x^2 dx \\
&= (n+1) \sum_{k=0}^n p_{n,k} \frac{1}{3} \left(\frac{(k+1)^3}{(n+1)^3} - \frac{k^3}{(n+1)^3} \right) = \\
&= \frac{1}{3} \sum_{k=0}^n p_{n,k} \frac{3k^2}{(n+1)^2} + \frac{1}{3} \sum_{k=0}^n p_{n,k} \frac{3k}{(n+1)^2} + \sum_{k=0}^n p_{n,k} \frac{1}{3(n+1)^2} = \\
&= \frac{n^2}{(n+1)^2} \sum_{k=0}^n \left(\frac{k}{n} \right)^2 p_{n,k}(x) + \frac{n}{(n+1)^2} \sum_{k=0}^n \frac{k}{n} p_{n,k}(x) + \frac{1}{3(n+1)^2} = \\
&= \frac{n^2}{(n+1)^2} B_n(e_2) + \frac{n}{(n+1)^2} B_n(e_1) + \frac{1}{3(n+1)^2} \xrightarrow{n \rightarrow \infty} 1e_2 + 0e_1 + 0
\end{aligned}$$

15. Să se arate că

$$\lim_{n \rightarrow \infty} B_n(e^x; x) = e^x,$$

pentru orice $x \in \mathbb{R}$, unde B_n este polinomul lui Bernstein.

Figure 1:

$$\begin{aligned} B_n(e^x)(x) &= \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} e^{\frac{k}{n}} = \\ &= \sum_{k=0}^n \binom{n}{k} \left(x e^{\frac{1}{n}}\right)^k (1-x)^{n-k} = \\ &= \left((1-x) + x e^{\frac{1}{n}}\right)^n = \\ &= \left(1 + x \left(e^{1/n} - 1\right)\right)^n = \\ &= \left[\left(1 + x \left(e^{1/n} - 1\right)\right)^{\frac{1}{x(e^{1/n}-1)}}\right]^{nx(e^{1/n}-1)} \rightarrow \\ &\xrightarrow{n \rightarrow \infty} e^{\lim_{n \rightarrow \infty} x \frac{e^{1/n} - 1}{1/n}} = e^x \end{aligned}$$

17. Se consideră șirul de operatori $(M_n)_{n \in \mathbb{N}}$, $M_n : C[0, 1] \rightarrow \Pi_n$, definiți prin:

$$M_n(f)(x) = (n+1) \sum_{k=0}^n p_{n,k}(x) \int_0^1 f(t) p_{n,k}(t) dt.$$

Să se calculeze

$$M_n(e_0), \quad M_n(e_1), \quad M_n(e_2)$$

și să se arate că

$$\lim_{n \rightarrow \infty} \|M_n(f) - f\|_{\infty} = 0.$$

Pt. $M_n(e_0)$ trebuie calculate:

$$\begin{aligned}
 I_{n,k} &= \int_0^1 \binom{n}{k} t^k (1-t)^{n-k} dt = \\
 &= \binom{n}{k} \int_0^1 t^k (1-t)^{n-k} dt = \\
 &= \binom{n}{k} B(k+1, n-k+1) = \\
 &= \binom{n}{k} \frac{\Gamma(k+1) \Gamma(n-k+1)}{\Gamma(k+1+n-k+1)} = \\
 &= \binom{n}{k} \frac{k! (n-k)!}{(n+1)!} = \frac{1}{n+1}
 \end{aligned}$$

24. Să se determine

$$\lim_{k \rightarrow \infty} B_{2k}(f; x),$$

unde

$$f(x) = \left[x + \frac{1}{2} \right].$$

Răspuns:

$$\lim_{k \rightarrow \infty} B_{2k}(f; x) = \begin{cases} 0, & x < 1/2 \\ 1/2, & x = 1/2 \\ 1, & x > 1/2 \end{cases}$$

1.1 O problemă de convergență

Exercise 1 Dacă $\lim_{n \rightarrow \infty} x_n = \alpha$ și $\lim_{n \rightarrow \infty} \frac{|x_{n+1} - \alpha|}{|x_n - \alpha|} = 0$ atunci:

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1} - x_n|}{|x_n - \alpha|} = 1.$$

Demonstrație: Ideea:

$$\frac{||\alpha - x_n| - |\alpha - x_{n+1}||}{|x_n - \alpha|} \leq \frac{|x_{n+1} - \alpha + \alpha - x_n|}{|x_n - \alpha|} \leq \frac{|x_{n+1} - \alpha| + |x_n - \alpha|}{|x_n - \alpha|}$$

de unde:

$$\left| 1 - \frac{|x_{n+1} - \alpha|}{|x_n - \alpha|} \right| \leq \frac{|x_{n+1} - x_n|}{|x_n - \alpha|} \leq \frac{|x_{n+1} - \alpha|}{|x_n - \alpha|} + 1$$

trecând la limită rezultă (cf. teoremei cleștelui) $\lim_{n \rightarrow \infty} \frac{|x_{n+1} - x_n|}{|x_n - \alpha|} = 1$.